AI FOR FOUNDATIONAL LEARNING HACKATHON

Saksham AI

A field coach for India's teacher mentors

Classroom evidence. Better conversations.
A clear next step for every visit.

CHALLENGE 5: SYSTEM CAPACITY



The mentoring gap between a visit and a change

CRPs are expected to help teachers improve classroom practice.

Our focus is the moment after an observation: choosing useful feedback, discussing it well, and returning to the agreed practice.

THE DOCUMENTED NEED

India's school supervision still gives substantial attention to compliance. The NITI Aayog report calls for stronger academic mentoring and CRP capacity.

It recommends regular classroom visits, follow-up support, and digital records of teaching practices.

Primary user: CRP / teacher mentor
Beneficiaries: teachers and early-grade learners

Sources: NITI Aayog, School Education System in India (2026), pp. 151–152; World Bank Coach (2021).

One observation becomes one usable coaching plan

01

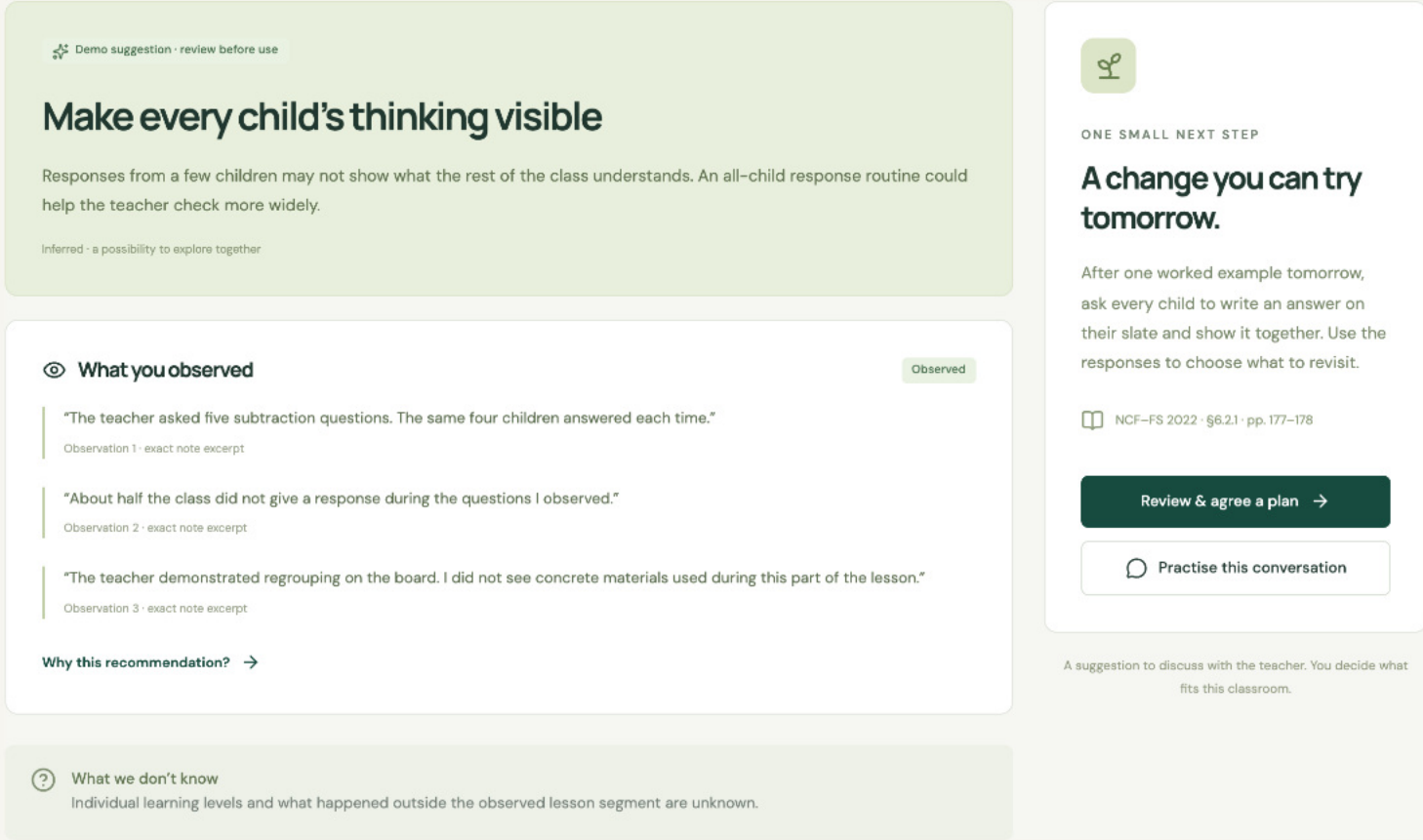
Capture evidence

Speak or type what happened during a short lesson segment.

02

Choose one next step

Review the interpretation, its source, and a practice the teacher can try.



Actual prototype capture with fictional sample observations. Demo output shown; mentor review remains required.

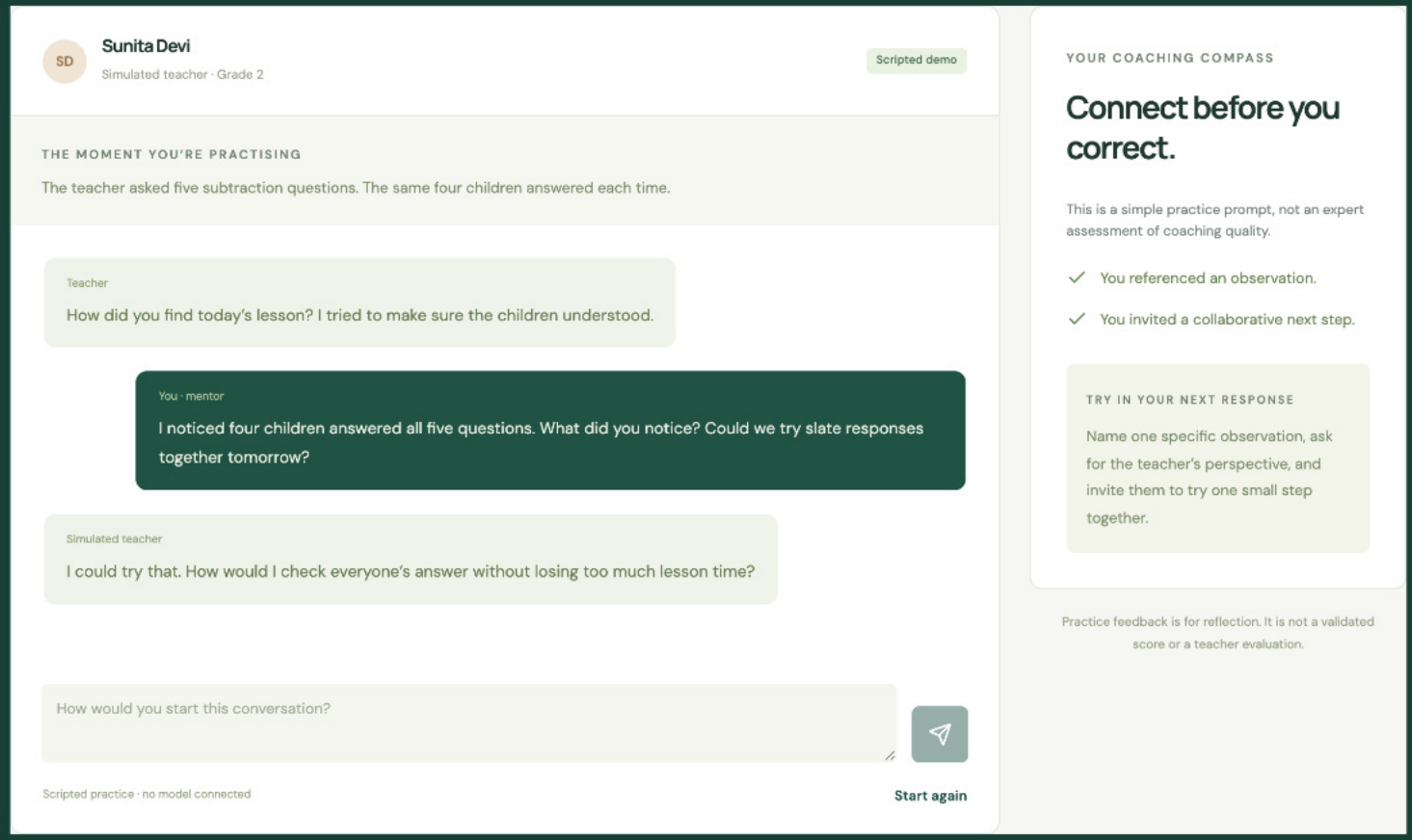
The mentor practises before the real conversation

AI takes the teacher's role

The mentor rehearses a response grounded in the observation.

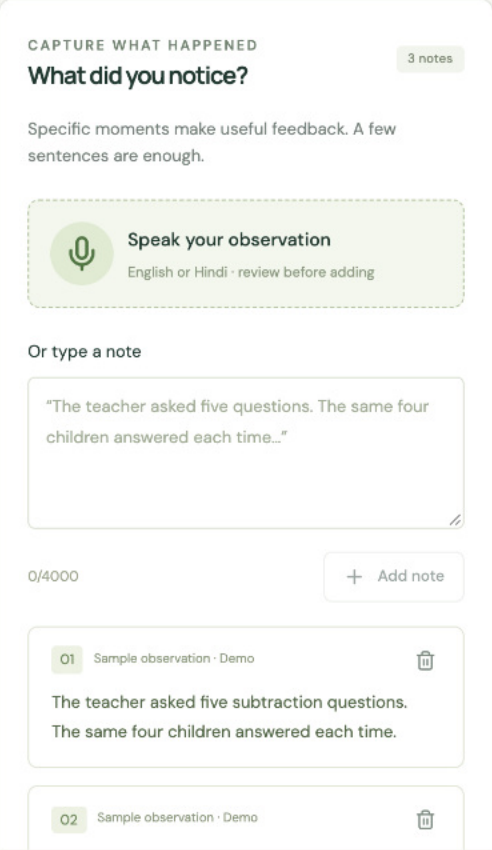
The coach gets feedback

On specificity, respectful inquiry, and a feasible next step.



Actual prototype capture. Scripted rehearsal shown. Adaptive Claude role-play is planned and requires validation.

Designed around the mentor’s field day



Meena, Cluster Resource Person

Supports early-grade teachers across a school cluster. Needs an observation she can use in a conversation, with minimal re-entry.

Field constraint	Product response
Limited time after a visit	A few notes and one coaching priority
Hindi or another local language	Hindi / English first; review the transcript
Intermittent connectivity	Device-local drafts; AI waits for connection
Trust and sensitive classroom data	Mentor notes, no continuous child recording

Meena is a fictional design persona. Constraints are design assumptions to validate with CRPs; voice support is browser-dependent.

A complete loop around one teaching practice

Prepare	Observe	Reflect	Rehearse	Commit
Review the last agreed practice	Capture short mentor notes	Check evidence and one next step	Practise the conversation	Agree an action and revisit date

At the next visit

"Did more children respond? What did the teacher change after seeing their answers?"

THE AGREED PRACTICE

After a worked example, every child writes an answer on a slate.

Mentor: co-plan the task and revisit in 14 days.

Illustrative Grade 2 scenario. A 14-day revisit is a proposed cadence, to be agreed locally; it is not an impact result.

A distinct CRP workflow within an existing field

Approach	Published focus	Saksham's proposed contribution
TeachFX	Lesson-audio analysis and teacher reflection	Mentor-authored field evidence, without requiring full-lesson audio
Mursion	Simulated conversations for educator skills	Rehearsal anchored to the current FLN observation and agreed action
DIKSHA / NISHTHA	Teacher professional learning resources	Relevant resource excerpts at the moment a mentor gives feedback

Our differentiation hypothesis

Hindi-first CRP support that connects evidence, coaching rehearsal and the next classroom visit.

Sources: TeachFX, Mursion and NCERT/DIKSHA public product pages. Comparison of stated focus, not an exhaustive feature audit.

Every recommendation has an evidence trail

OBSERVED

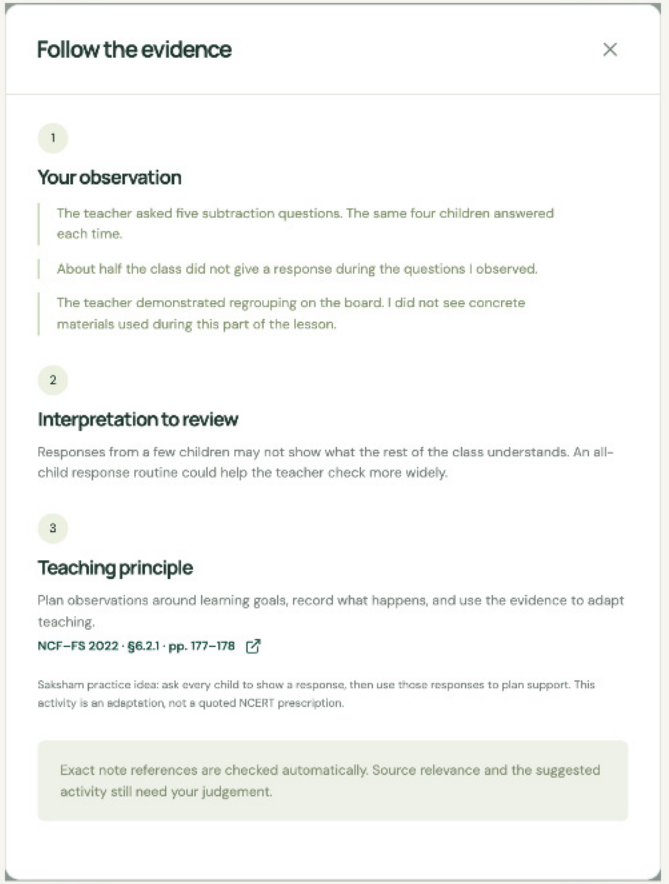
“The same four children answered.”

INFERRED

The teacher may need a way to see what the rest of the class understands.

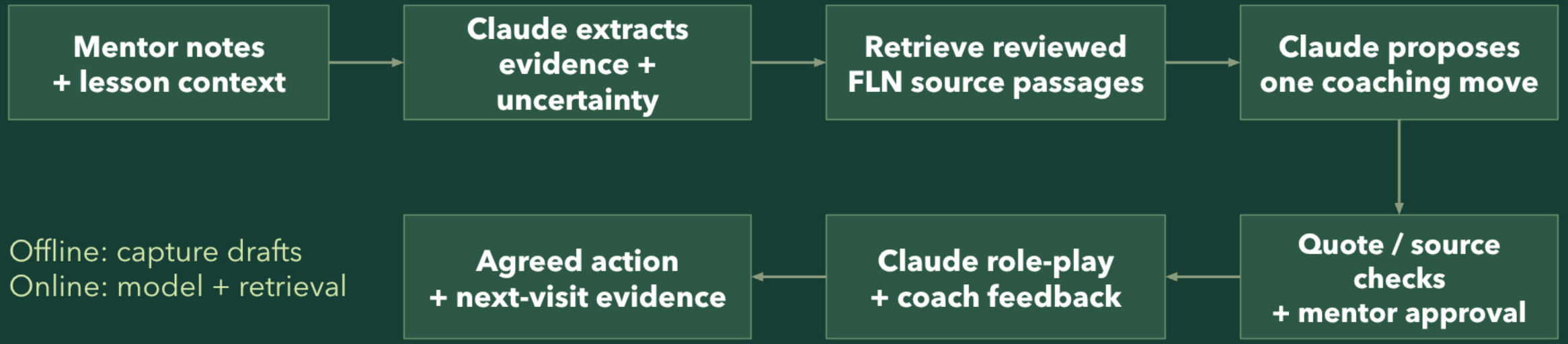
UNKNOWN

Individual learning levels and what happened outside this lesson segment.



Source principle: NCERT NCF-FS 2022, §6.2.1, pp. 177–178. The suggested activity is a Saksham adaptation.

AI reasons over evidence before a mentor approves



DATA

NCF–FS, NIPUN Bharat and permitted state FLN resources. Expert-reviewed scenarios for evaluation. No fine-tuning planned.

IMPLEMENTATION

Next.js / TypeScript, server-side Claude, schema validation. Planned: PostgreSQL + pgvector, secure sync and multilingual speech.

Proposed production flow. Prototype uses fixed curated summaries; document retrieval, sync and multilingual ASR are build milestones.

A measurable path to a field pilot

What we will test

Check	Proposed acceptance target
Evidence precision	≥95% of cited claims supported
Expert agreement	≥80% agreement on top coaching priority
Mentor usability	Median reviewed plan in ≤3 minutes

40 synthetic cases, reviewed by FLN experts.
Then 5 consenting mentors across 10 paired visits.
Measure practice adoption separately from learning gains.

BUILD MILESTONES

By 27 September

Submit the proposal and demonstrate the core interaction.

If shortlisted: by 1 November

Validate live AI and source retrieval. Test Hindi use and device constraints. Publish the testable build and demo.

All sample sizes and thresholds below are proposed validation targets, not measured results. No pilot partners are confirmed.